OSSP (Operating Systems And Systems Programming) - 25CS2104E

Roll No. : 2520030106

Name: D Mohit Venkat Sai

SEC-8

SKILL-7:

Aim: Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

PROCEDURE :

1. **Compilation :**

```
dmvs@DMVS:~/skill7$ mkdir skill7
dmvs@DMVS:~/skill7$ cd skill7
dmvs@DMVS:~/skill7$ nano path_search.c
dmvs@DMVS:~/skill7$ gcc path_search.c -o path_search
dmvs@DMVS:~/skill7$ ./path_search ls
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/
wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common F
iles/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt
/c/Windows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program
Files/Git/cmd:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/
Users/dmvs/.local/bin:/mnt/c/Users/dmvs/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/dmvs/AppD
ata/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dmvs/Downloads/jdk-21_windows-x64_bin.exe:/m
nt/c/Users/dmvs/AppData/Roaming/npm:/snap/bin

Command found: /usr/bin/ls
Parent process waiting for child...

Child process started. PID = 1804
path_search  path_search.c
Child terminated with exit status: 0
Parent process completed.
```

2. **Valid command :**

```
dmvs@DMVS:~/skill7$ ./path_search pwd
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/
wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common F
iles/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt
/c/Windows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program
Files/Git/cmd:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/
Users/dmvs/.local/bin:/mnt/c/Users/dmvs/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/dmvs/AppD
ata/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dmvs/Downloads/jdk-21_windows-x64_bin.exe:/m
nt/c/Users/dmvs/AppData/Roaming/npm:/snap/bin

Command found: /usr/bin/pwd
Parent process waiting for child...

Child process started. PID = 1816
/home/dmvs/skill7
Child terminated with exit status: 0
Parent process completed.
```

3. Missing command :

```
dmvs@DMVS:~/skill7$ ./path_search hello123
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/
wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common F
iles/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt
/c/Windows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program
Files/Git/cmd:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/
Users/dmvs/.local/bin:/mnt/c/Users/dmvs/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/dmvs/AppD
ata/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dmvs/Downloads/jdk-21_windows-x64_bin.exe:/m
nt/c/Users/dmvs/AppData/Roaming/npm:/snap/bin

Command 'hello123' not found in PATH.
dmvs@DMVS:~/skill7$ ./path_search cat
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/
wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common F
iles/Oracle/Java/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt
/c/Windows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program
Files/Git/cmd:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/
Users/dmvs/.local/bin:/mnt/c/Users/dmvs/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/dmvs/AppD
ata/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/dmvs/Downloads/jdk-21_windows-x64_bin.exe:/m
nt/c/Users/dmvs/AppData/Roaming/npm:/snap/bin

Command found: /usr/bin/cat
Parent process waiting for child...

Child process started. PID = 1843
Parent process completed.
```

Code:

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/stat.h>

int main(int argc, char *argv[])
{
    if (argc != 2)
    {
        printf("Usage: %s <command>\n", argv[0]);
        return 1;
    }

    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH Variable:\n%s\n\n", path);

    char *path_copy = strdup(path);
    char *dir = strtok(path_copy, ":");

    int found = 0;

    while (dir != NULL)
    {
        char fullpath[1024];

        snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, arg>
```

```
        struct stat st;

        if (stat(fullpath, &st) == 0)
        {
            if (access(fullpath, X_OK) == 0)
            {
                printf("Command found: %s\n", fullpath);
                found = 1;
                break;
            }
            else
            {
                printf("Found but not executable: %s\n", fullp>
            }
        }

        dir = strtok(NULL, ":");
    }

    free(path_copy);

    if (!found)
    {
        printf("Command '%s' not found in PATH.\n", argv[1]);
        return 1;
    }

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
```

```
    if (pid == 0)
    {
        printf("\nChild process started. PID = %d\n", getpid()>

        execlp(argv[1], argv[1], (char *)NULL);

        perror("exec failed");
        exit(1);
    }
    else
    {
        int status;

        printf("Parent process waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status))
        {
            printf("Child terminated with exit status: %d\n",
                   WEXITSTATUS(status));
        }

        printf("Parent process completed.\n");
    }

    return 0;
}
```

Observation :

- The PATH environment variable was retrieved successfully.
- The PATH was divided into separate directories and searched.
- The required executable command was located successfully.
- Execute permission was verified using access().
- A child process was created using fork().
- The parent process waited for the child using waitpid().
- Missing commands were detected and an error message was displayed.

Result :

The program successfully implemented **process synchronization using waitpid()** and **command resolution using the PATH variable**. It correctly searched for executables, verified permissions, executed valid commands, monitored the child process, and handled missing commands successfully.